

PICTURE SLIDESHOW Using ESP32

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A small TFT display connected to Raspberry Pi can display many pictures one after another. This can also be done using ESP32, using the same calculations and tricks. This project is used to create a picture slideshow using ESP32 and an SD card shield. The display can be ILI9163 TFT, Nokia 5110 LCD or equivalent. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of ESP32 picture slideshow is shown in Fig. 2. Main components used in this project are ESP32 module, ILI9163 TFT display module, SD card shield and SD card. Details of the connections of the modules and shield are given in the table.

ESP32. It is a low-power micro-

controller (MCU) with integrated Wi-Fi and dual-mode Bluetooth—it is a successor of ESP8266 MCU. Once programmed, ESP32 never hangs, because it works on 3.3 volts and a single set of instructions, unlike Raspberry Pi. ESP32 is easy to handle and works well for small programs.

Pin map diagram of SparkFun ESP32 is shown in Fig. 3. The modified library and other associated files are packed in a folder for this project. Arduino IDE is used to upload the sketch to ESP32. (ESP-Wroom-32 version 1.1 was used at EFY Lab for testing purposes.)

TFT display. The 3.65cm (1.44-inch) coloured TFT screen display with SPI serial bus was used during testing. It is compatible with Nokia 5110 display, and is available on www.aliexpress.com

SD card shield. This requires a 5V DC power supply or smartphone charger. There are many SD cards

available online, but 6-pin connector SD card shield was used at EFY Lab. It is available on www.ebay.in

Software

Arduino sketch `esp_image.ino` is simple to understand. The libraries required in the sketch include `TFT_ILI9163C`, `Adafruit_GFX_Library`

CONNECTIONS BETWEEN MODULES AND SHIELD

ILI9163 TFT Module Pin	ESP32 Module Pin	SD Card Shield
LED/Vcc	3.3V	3.3V
GND	GND	GND
A0	P15	P19
CS	P2	P22
Reset	P4	
SCK	P18	P18
SDA	P23	P23
		MOSI

*Some SD card shields may not work on 3.3V. In that case use a 5V for Vcc.



Fig. 1: Author's prototype

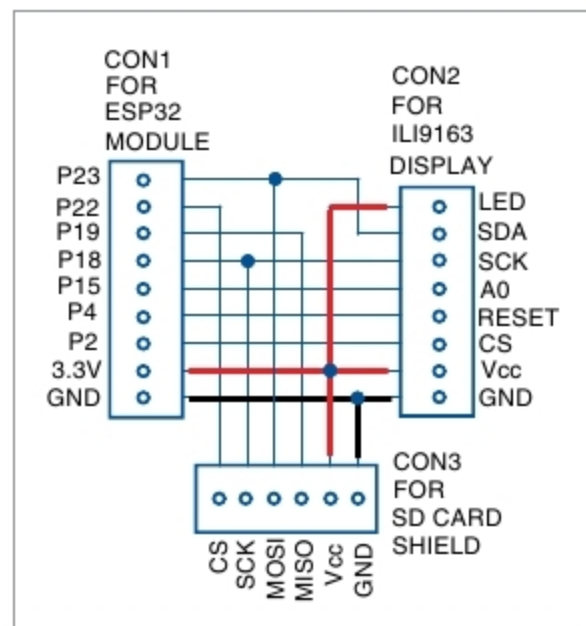


Fig. 2: Circuit diagram of ESP32 picture slideshow

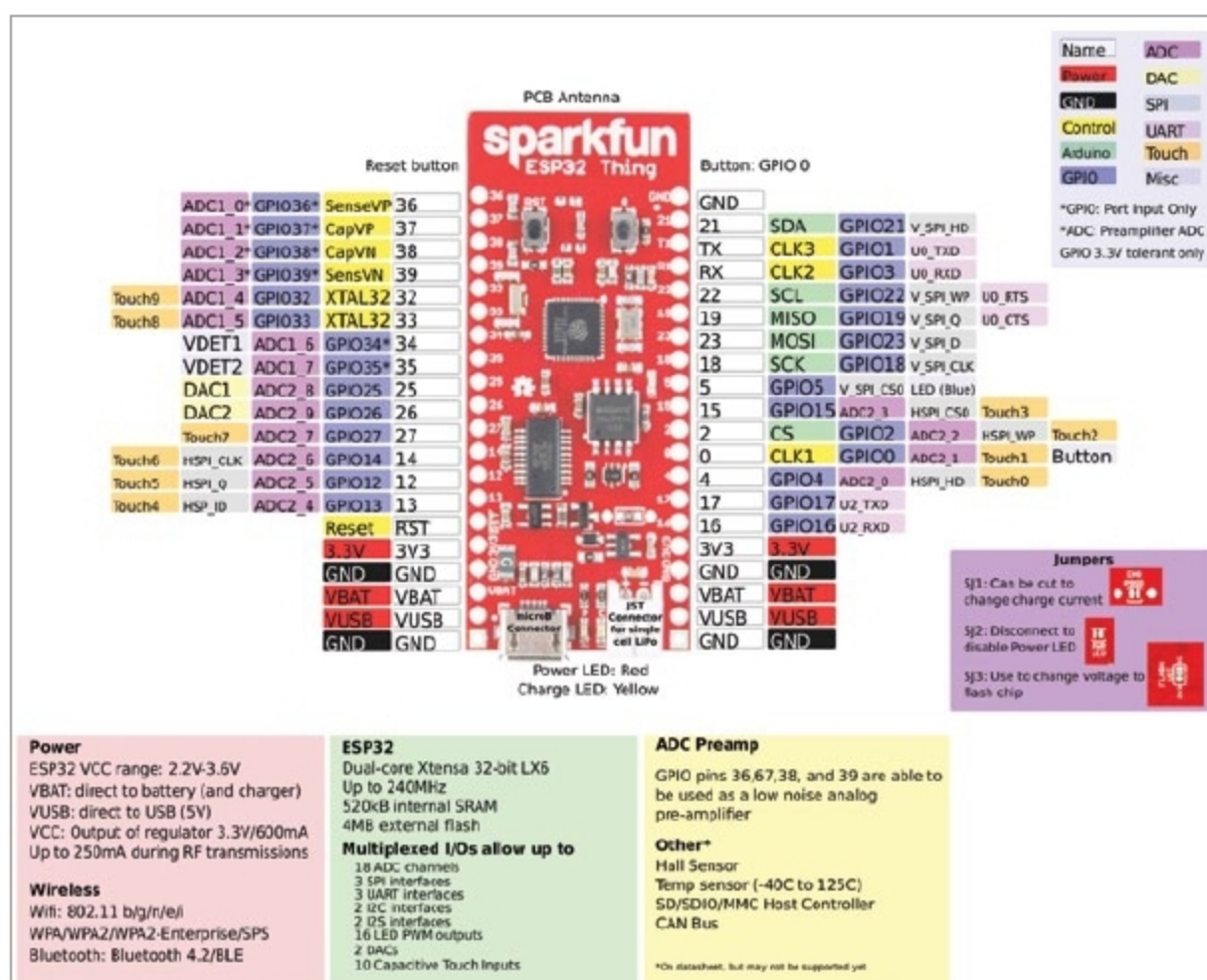


Fig. 3: ESP32 pin details